SOLUTION
SECTION-I

Single Correct Answer Type **3 Q. [3 M (–1)]**

1. Ans. (B)

Sol. $V_g = 10 \times 50\mu A$

$$R_V = \left(\frac{V}{V_g} - 1\right) R_g$$

$$R_A = \frac{R_g}{\left(\frac{I}{I_g} - 1\right)}$$

2. Ans. (C)

3. Ans. (C)

Multiple Correct Answer Type **7 Q. [4 M (–1)]**

4. Ans. (A,D)

Sol. $i_R = 2A$

$\therefore i_{R_1} = i_{R_2} = 1A$

$V_{R_1} = 30V = 1 \times R_1$

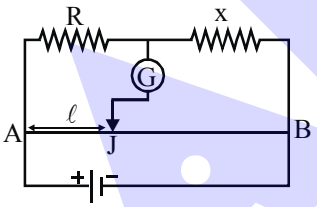
$\Rightarrow R_1 = 30\Omega$

$V_{R_2} = 40 - 30 = 10 = 1 \times R_2$

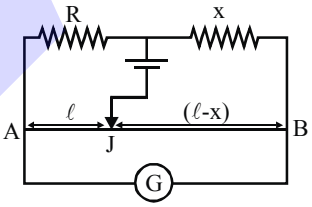
$\Rightarrow R_2 = 10\ \Omega$

5. Ans. (AC)

Sol. (i)



$$\frac{R}{x} = \frac{\ell}{\ell_0 - \ell} \quad \ldots (i)$$



Here $V_A = V_B \Rightarrow$ current through galvanometer is zero.

$\frac{R}{\ell} = \frac{x}{\ell_0 - \ell}$ [it is also a balance Wheatstone bridge]

(iii) Ratio $\frac{R}{x} = \frac{\ell}{\ell_0 - \ell}$ is independent of cross-section area and resistivity

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(iv) $\frac{\ell}{\ell_0-\ell}=\frac{R}{x}\Rightarrow\frac{2x}{2R}\neq\frac{\ell}{\ell_0-\ell}$

6. Ans. (B,C)

Sol. In the standard PO experiment, reading are taken such that the galvanometer deflects in opposite direction. The key connected to galvanometer is switched on after the key of the battery so as to reduce the inductive effect in the circuit while taking the reading. The same is true while switching off the setup.
Note : Refer the PO Box experiment.

7. Ans. (C, D)

Sol. $i=\frac{3}{30}=\frac{1}{10}$

$v_{AB}=\frac{1}{10}\times 20=2v$ = maximum voltage read by potentiometer

also voltage per mm $=\frac{2}{20\times100}=10^{-4}v=0.1mv$

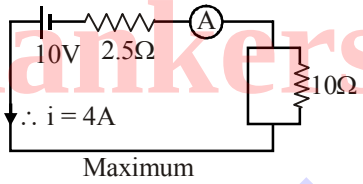
$\Rightarrow$ accuracy of potentiometer is 0.1 mv

8. Ans. (A,C)

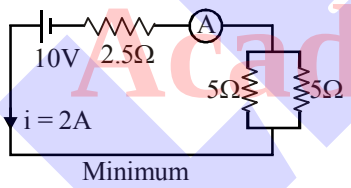
Sol. i = 3A
and V = iR = (1) (50) = 50 V

9. Ans. (B,C,D)

Sol.



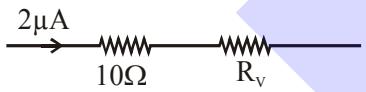
Maximum



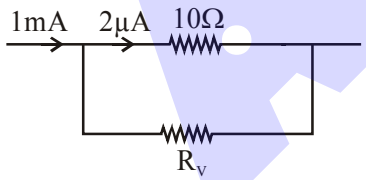
Minimum

10. Ans. (A,C)

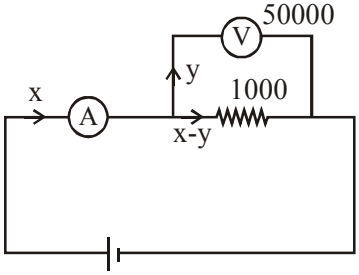
Sol.



$0.1=2\times10^{-6}(10+R_V)$
$\therefore R_V=49990\ \Omega$



$2\times10^{-6}\times10=10^{-3}R_A\ \therefore R_A=0.02\ \Omega$



y 50000 = (x – y) 1000

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$\therefore$ $51y = x$

$$\text{Reading} = \frac{y\,50000}{x} \simeq 980$$

Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

11. Ans. (C)

Sol. $i = \dfrac{\varepsilon}{R_g + R}$

12. Ans. (C)

Sol. $10 \times 10^{-3} = \dfrac{1}{20 + R} \quad \Rightarrow R = 80\Omega$

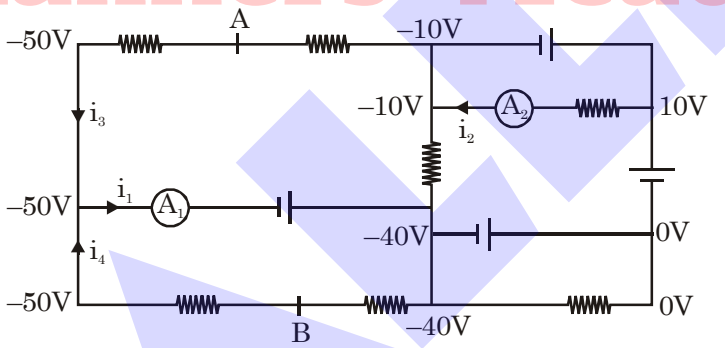
13. Ans. (A)

Sol. $I = \dfrac{1}{20 + 80 + 100} = 5\text{ mA}$

Matching list based comprehension Type (4 × 4 × 4) 1 Table × 3 Q. [3(–1)]
Single option correct (Three Columns and Four Rows)

14. Ans. (B)
15. Ans. (C)
16. Ans. (D)
Sol. (14 to 16)

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$V_2 = 40$ V when S_4 opened and $V_2 = 45$ V otherwise
$V_1 = 50$ V when S_3 opened and $V_1 = 30$ V otherwise
$A_2 = 0$A when S_1 opened and $A_2 = 4$A otherwise
$A_1 = 5$A when S_3 and S_4 are closed (no effect of S_1 and S_2)
$A_1 = 4$A when S_4 opened
$A_1 = 1$A when S_3 opened

	V_1	V_2	A_1	A_2	$\lvert A_1 - A_2 \rvert$
S_1opened	30	45	5	0	5
S_2opened	30	45	5	4	1
S_3opened	50	45	1	4	3
S_4opened	30	40	4	4	0

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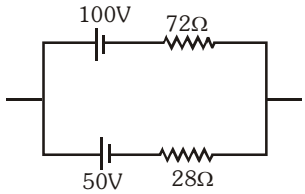
SECTION-II

Numerical Answer Type Question **4Q.[3(0)]**

(upto second decimal place)

17. Ans. 80.00

Sol.



$$I = \frac{50}{100} = 0.5 \text{ , } \phi = \frac{V}{100} = \frac{IR}{100} = \frac{800}{100} \times \frac{100}{1000} \text{ V/cm}$$

$\Delta V = \phi \ell_1$

Now $100 - 72 \times \frac{1}{2} = \ell_1 \times \frac{100}{1000} \times \frac{800}{100} \Rightarrow \ell_1 = 80\text{cm}$

18. Ans. 48.00

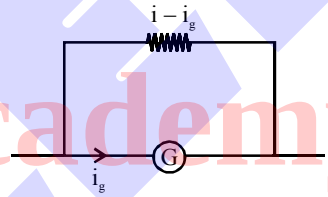
Sol. The current i in the galvanometer is directly proportional to the angle of deflection ($i \propto \theta$)

Then, $\frac{i_g}{i} = \frac{12}{60} = \frac{1}{5}$

or, $i_g = \frac{i}{5}$ .....(i)

For shunted galvanometer,

$(i - i_g)S = i_g G$



or, $G = (i - i_g)\frac{S}{i_g}$ ..... (ii)

Putting $i_g = \frac{i}{5}$ from equation (i) in equation (ii) and S = 12 ohm,

G = 48 ohm

19. Ans. 40.00

Sol. In **case–1**

'i' is very small $\Rightarrow R_V$ is very high

In **case - 2**

V across ammeter is V, small $\Rightarrow R_A$ is very low

$\Rightarrow$ from case - 1, we can say that

$V_{Cell} = 100$ volt.

From case 2, we can say

$25 \times 10^{-3} = V_{cell} - ir$

$= 100 - 2.5\ r$

$r = 40\ \Omega$

20. Ans. 35.00

Sol. Total resistance $= \int_0^L \lambda x\, dx = \frac{\lambda}{2} L^2$

For balance condition

$$\frac{R_1}{R_2} = \frac{(\lambda/2) l_1^2}{(\lambda/2)(L^2 - l_1^2)}$$

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Similarly, $\frac{R_2}{R_1} = \frac{l_2^2}{L^2 - l_2^2}$

$\therefore \quad \frac{R_1}{R_2} = \frac{l_1^2}{L^2 - l_1^2} = \frac{L^2 - l_2^2}{l_2^2} = n$

Solving, $l_1 = L\sqrt{\frac{n}{n+1}}$ and $l_2 = L\sqrt{\frac{1}{n+1}}$

$\therefore \quad \Delta = l_1 - l_2 = \frac{L}{\sqrt{n+1}}(\sqrt{n} - 1)$

Putting L = 100 cm and n = 3

$\Delta = \frac{100}{2}(\sqrt{3} - 1) = 35$ cm

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

21. Ans. 2

Sol. Let x be the resistance between a & b then, x || r.

The total resistance of circuit $= \frac{rx}{r+x} + (R - x)$

So, current through ammeter $I_A = \frac{I_0 r}{r+x} = \frac{V_0 r}{rx + (R-x)(r+x)}$

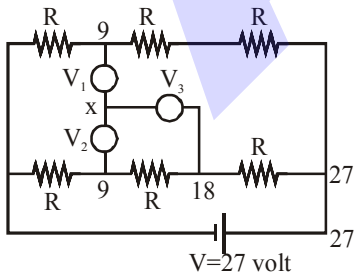
for its minimum value,

$\left(\frac{dI_A}{dx}\right) = 0 \Rightarrow$ we get $x = \frac{R}{2}$

So, $(I_A)_{min} = \frac{V_0 r}{R\left(r + \frac{R}{4}\right)} = \frac{(12)(2)}{4(2+1)} = 2A$

22. Ans. 6

Sol.



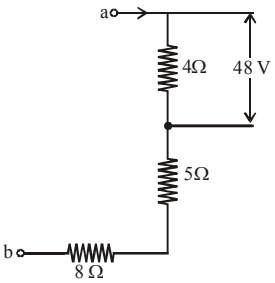
$\frac{x-9}{R} + \frac{x-9}{R} + \frac{x-18}{R} = 0 \Rightarrow x = 12 \quad \therefore V_3 = 6$ volt

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23. Ans. 4

Sol.



$\Rightarrow i = 12$

& $V_{ab} = 17 \times 12 = 204$ Volt

24. **Ans. 2**

Sol. $\frac{x}{R} = \frac{\ell}{1-\ell} \Rightarrow x = \frac{40}{60} \times 50 = \frac{100}{3}\Omega$

$$\frac{\Delta x}{x} = \frac{\Delta \ell}{\ell(1-\ell)} \Rightarrow \frac{\Delta x}{x} = \frac{0.1}{40 \times 60} = \frac{1}{240}\%$$

$$x = \frac{\rho \ell}{\frac{\pi d^2}{4}} \Rightarrow \rho = \frac{\pi x d^2}{4\ell}$$

$$\Rightarrow \frac{\Delta \rho}{\rho} = \frac{\Delta x}{x} + \frac{2\Delta d}{d} + \frac{\Delta \ell}{\ell}$$

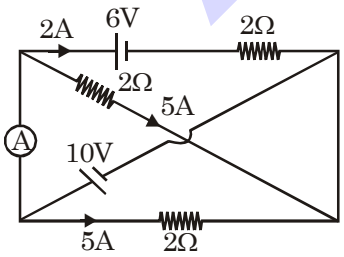
$$= \frac{1}{240} + 2 \times \frac{0.01}{1.6} \times 100 + \frac{0.01}{2.25} \times 100$$

$$\frac{\Delta \rho}{\rho} = \frac{2}{1.6} + \frac{1}{4.25} = 1.25 + 0.44 + 0.0042$$

$= 1.69 \approx 2\%$

25. **Ans. 7**

Sol. From diagram



Current through ammeter

$= (2 + 5) = 7A$

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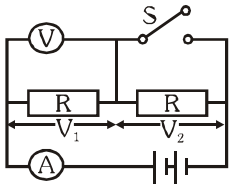
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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

26. Ans. (A) → (P) ; (B) → (P) ; (C) → (Q) ; (D) → (Q,S)

Sol.



After closing the switch net resistance decreases therefore there will be increases in the current.

Initially $V = V_1+V_2$ after closing the switch V_2 becomes zero hence $V = V_1$.

After short circuiting current in the resistance becomes zero therefore power become zero.

Subjective Type **4 Q. [4 M (0)]**

27. **Ans.** First case $V_1 > V_2 > V_3$; Second case $V_1 = V_2 = V_3$

Sol. $I = \frac{6}{R_{eq}}$, $V_{1k\Omega} = I \times 1$, $V_{2k\Omega} = 6 - V_{1\,k\Omega}$

(a) $R_{eq} = 1 + \frac{2\times 30}{2+30} = \frac{23}{8}k\Omega$

$$I = \frac{6}{\left(\frac{23}{8}\right)} = \frac{48}{23}mA,\ V_{1k\Omega} = \frac{48}{23}V$$

$$V_{2k\Omega} = 6 - \frac{48}{23} = \frac{90}{23}V = 3.91\ V$$

(b) $R_{eq} = 1 + \frac{2\times 1}{2+1} = \frac{5}{3}k\Omega$

$$I = \frac{6}{\left(\frac{5}{3}\right)} = \frac{18}{5}mA,\ V_{1k\Omega} = \frac{18}{5}V$$

$$V_{2k\Omega} = 6 - \frac{18}{5} = \frac{12}{5}V = 2.4\ V$$

(c) $R_{eq} = 1 + \frac{1}{\frac{1}{1}+\frac{1}{2}+\frac{1}{30}} = \frac{38}{23}k\Omega$

$$I = \frac{6}{\left(\frac{38}{23}\right)} = \frac{138}{38}mA,\ V_{1k\Omega} = \frac{138}{38}V$$

$$V_{2k\Omega} = 6 - \frac{138}{38} = \frac{90}{38}V = 2.37\ V$$

$[V_1 > V_2 > V_3]$

In second case

$V_A - V_B = 6\ V$

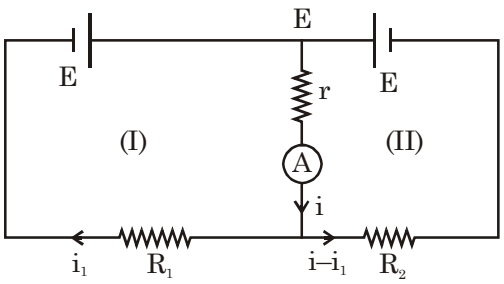
$V_1 = V_2 = V_3$

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28. **Ans.** $\frac{\varepsilon}{r+R_0(f-f^2)}$; for I_{max}, $f = 0, 1$; I_{min}, $f = 1/2$

Sol.



$R_1 = fR_0$ $\quad R_2 = (1-f)R_0$

I[st] Loop

$E - ir - fi_1R_0 = 0$

II[nd] Loop

$E - ir - (i-i_1)R_0(1-f) = 0$

$fi_1R_0 = (i-i_1)R_0(1-f)$

$i_1 = i(1-f)$

$\therefore \quad E - ir - f(1-f)iR_0 = 0$

$$i = \frac{E}{r+R_0(f-f^2)}$$

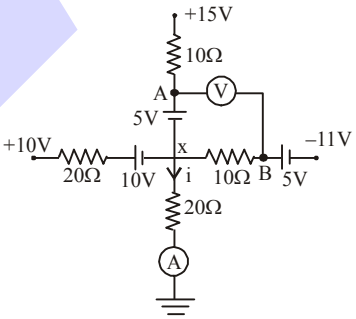
For $f = 0,1$ I will be maximum

For $f = 1/2$ I will be minimum

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29. **Ans.** $x = \frac{4}{3}V,\ 12\frac{1}{3}V,\ \frac{1}{15}A$

Sol. $\frac{x+10-10}{20}+\frac{x+5-15}{10}+\frac{x-(-6)}{10}+\frac{x-0}{20}=0$



$\frac{3x}{10}=1-\frac{6}{10}$

$x=\frac{4}{3}V$

$V_A = x+5=\frac{4}{3}+5=\frac{19}{3}V$

$V_B = -11+5=-6$

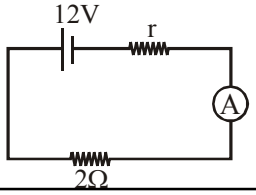
$V_A - V_B = \frac{19}{3}+6=\frac{37}{3}=12\frac{1}{3}V$

$i=\frac{x-0}{20}=\frac{4}{3\times 20}=\frac{1}{15}A$

30. **Ans.** (i) 1.01Ω (ii) 0-5 A, 0-10V, (ii) 0.05 A

Sol. (i) $r_A = \frac{1\times 99}{100}=0.99A$

$$3=\frac{12}{r+2+0.99}$$



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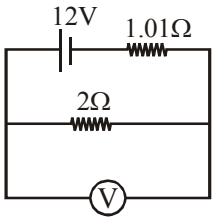
$\Rightarrow$ r + 2.99 = 4

r = 1.01 Ω

(ii) r_v = 99 + 101 = 200 W

$$i = \frac{12}{1.01 + \frac{2 \times 200}{200}} = 4A$$

V = 8V



$$\text{Range of voltmeter} = \frac{5}{4} \times 8 = 10V$$

Range of ammeter = 5A

(ii) $1(5 - i_g) = 99\, i_g$

$5 = 100\, i_g$

$i_g = 0.05$ A

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